

Production Scheduling Under Demand Changes

Create a production schedule that meets changing demand while respecting machine capacity and due dates.

Customer AeroSupply Operations	Department Production Planning	Difficulty Advanced	Time 2 class periods
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Production Goal: Meet demand with limited machine capacity.

Background

AeroSupply receives urgent orders for three component types. Each product requires different processing time and setup. Your team must build a schedule that minimizes late orders and unnecessary changeovers.

Learning Objectives

- Translate demand into capacity requirements
- Create and evaluate production schedules
- Consider setup time and bottlenecks
- Use metrics to justify a schedule

Equipment

- Demand/order table
- Machine capacity data
- Processing and setup-time table
- Spreadsheet or scheduling board
- Engineering notebook

Student Instructions

- 1 Calculate total required processing time for each product.
- 2 Determine available capacity for each shift or day.
- 3 Identify the bottleneck resource.
- 4 Create Schedule A that prioritizes earliest due date.
- 5 Create Schedule B that groups products to reduce setup time.
- 6 Calculate late orders, total setup time, and machine utilization for both.
- 7 Create a third hybrid schedule that improves the weak metric.
- 8 Recommend the best schedule and explain the customer tradeoff.

Engineering Requirements

- Schedule every order
- Do not exceed available machine capacity
- Report lateness, setup time, and utilization
- Meet all high-priority aerospace orders if feasible

Constraints & Safety

- Use the provided processing times

- Machines cannot process two jobs simultaneously
- Setup time must be included
- Document any order that cannot be completed on time

Deliverables

- Capacity calculations
- Three schedules
- Metric comparison table
- Recommended schedule
- Customer-impact explanation
- Reflection

Reflection Questions

- 1 Why did reducing setups sometimes delay an order?
- 2 Which resource limited output?
- 3 How would overtime change the plan?
- 4 What data would improve the schedule?

Engineering Support

Engineering Tip

A schedule is only feasible if every processing and setup minute fits within available capacity.

Common Mistake

Sorting orders by due date without including setup time can create an impossible schedule.

Stretch Goal

Add a machine breakdown and revise the schedule within ten minutes.

Career Connection

Production planners balance customer due dates, equipment capacity, labor, and setup losses.

Before You Submit

- Notebook evidence included
- Requirements checked
- Testing documented
- Reflection completed